Object Detection Part-5: Introduction to object detection

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In the last make of computer vision, Object Detection takes center stage, allowing computers to understand images in a way that goes beyond simple pattern recognition. This blog delves into the fascinating world of object detection, exploring its purpose their exact location within an image, how self-driving cars and security cameras in healthcare tools and your self-checkout lane at the grocery store, Object Detection is everywhere. It helps our pets recognize, keeps an eye on who's coming and going, assists in medical diagnoses, and even helps us navigate through the store. Let's take a journey into the basics of Object Detection.

Object Localization:

In order to build an object detector, you first learn about object localization. Let's start by defining what that means. Traditionally, image classification involved training computers to recognize and categorize objects within images. A machine learning model, typically a Convolutional Neural Network (CNN), would be trained on a dataset containing images of specific objects. Object localization takes this a step further by not only recognizing objects but also precisely locating them within an image. Imagine the shift from saying, "This is a cat," to saying, "This is a cat, and it's right there in the bottom left corner of the picture." This enhanced capability opens up a wealth of possibilities in various fields, ranging from understanding the spatial arrangement of objects in a scene.

1. Localization: Essentially drawing boxes around the objects. This helps the computer understand where things are in a picture.
2. Classification: Beyond just seeing, "Hey, there's a thing here," Object Detection also labels each object. So, if there's a car in the image, it doesn't just know it is there, it knows what exactly is that i.e. a car.

At the heart of object localization lie four key parameters:

1. Midpoint (bx, by): Representing the center of the bounding box, these coordinates pinpoint the object's location in the image.
2. Height and Width (bh, bw): These parameters dictate the dimensions of the bounding box, outlining the object's size within the image.

To teach a machine to localize objects, we introduce additional labels. These include:

1. **Object Probability (pc):** A binary indicator (1 or 0) denoting whether the object or objects of interest is present in the image.
2. **Bounding Box Parameters:** If an object is present (pc = 1), the machine learns the specific coordinates (bx, by) and dimensions (bh, bw) of the bounding box.
3. **Class Labels:** Indicating the category of the object (e.g., car, pedestrian), these labels provide a comprehensive understanding of the object's identity.

Let's dive into the scenario where we're developing a model to localize pedestrians, motorcycle in an image. In this context, our goal is not only to identify that there is an object in the image but also to precisely locate it within the image frame. The first step involves defining the output components of our model.

Probability of Object (pc): This component signifies whether an object is present in the image. If a pedestrian, motorcycle, or any other object from our predefined classes (Cars and motorcycles in Class 2, 3) of these objects are detected, pc will be equal to 1; otherwise, it will be 0. Here, we have a car, So, pc = 1.

Bounding Box Parameters (bx,by,bh,bw): These parameters define the bounding box around the detected object. They include the horizontal position of the bounding box midpoint (bx), the vertical position (by), the height normalized to the image height (bh), and the width normalized to the image width (bw).

Class Probabilities (c1,c2,c3): When an object is present (pc=1), these components provide probabilities for each class (pedestrian, car, motorcycle).

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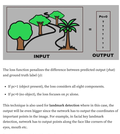
We can interpret this as:

- If pc=1, the model indicates the presence of an object, and the bounding box parameters (bx,by,bh,bw) precisely locate the object. The class probabilities (c1,c2,c3) convey the likelihood of the object belonging to each class.

Conceptual Image

- If pc=0, no specified objects are present in the image, and the remaining parameters become "don't cares."



The loss function penalizes the difference between predicted output (*pred*) and ground truth label (*y*):

- If pc=1 (object present), the loss considers all eight components.
- If pc=0 (no object), the loss focuses on pc alone.

This technique is also used for **landmark detection** where in this case, the output will be even bigger since the network has to output the coordinates of multiple landmarks on the image. For example, in facial landmark detection, network has to output points along the face like corners of the eyes, mouth etc.

Basic structure of object localization networks:

In an object localization task, the goal is to detect and localize objects within an image. The network typically consists of two main components: a backbone and optionally have one parts. An encoder or a feature extractor such as a backbone, takes an image as input and extracts relevant features used to locate and label objects. The output features from the extractor are passed to a decoder which predicts the bounding boxes and labels. After the feature extraction, the network splits into two heads, each with specialized roles. The first head is the localization head, responsible for predicting the bounding box coordinates. This involves regression, where the regression head outputs a vector that represents the bounding box coordinates. The vector contains information about the position and size of the bounding box in the image. The second head is the classification head, responsible for predicting the class probabilities for each object class. This is essentially a multi-class classification task where the network assigns a probability distribution across different classes to each bounding box. The output of the classification head is a vector with probabilities for each class. These probabilities are computed using a softmax activation function, ensuring that the sum of probabilities across all classes equals one. The class with the highest probability is typically chosen as the predicted class for the corresponding bounding box. During training, the network learns to adjust its parameters based on the provided ground truth bounding boxes and class labels. The goal is to minimize the overall loss, which encourages accurate localization and correct class predictions.

Object Localization vs Object Detection:

Object detection introduces two key challenges: **classification with localization** and the more complex **detection** problem. In classification with localization or object localization, the algorithm not only identifies the object in an image but also draws a bounding box around it, precisely drawing a bounding box around it. This process, known as localization, is crucial for understanding where the recognized object is positioned in the image. However, object detection takes this further by considering the number of objects in existence or not within an image. Consider scenarios with three cats in an image. Using object localization approach, we would need to predefine three bounding boxes to cover each cat individually. *What if you look at the output vector of object localization, you can see there is only one set of bounding box coordinates and one set of class labels. So we need to define 3 sets of bounding box coordinates and the model must learn to* Unsplash
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